

Real-world Data Project

Health Domain: End-to-End Data Analysis and Prediction

Project Overview

This document contains a complete, working Python script designed to fulfill the requirements of the **Real-world Data Project** assignment. The project focuses on the Health domain, utilizing patient records to perform end-to-end data analysis, build a predictive model, and present findings visually.

Prerequisites

Before running the code, ensure you have the required libraries installed. You can install them via your terminal or command prompt:

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

Python Source Code

The following script uses the built-in Diabetes Dataset from the `scikit-learn` library. This ensures the code runs flawlessly without requiring external CSV downloads.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

# =====
# 1. CHOOSE A DATASET (Health / Patient Records)
# =====
print("Loading Diabetes Patient Dataset...")
diabetes = load_diabetes()
df = pd.DataFrame(diabetes.data, columns=diabetes.feature_names)

# The target is a quantitative measure of disease progression one year after baseline
df['target_progression'] = diabetes.target
```

```

print("\n--- First 5 rows of the dataset ---")
print(df.head())

# =====
# 2. END-TO-END DATA ANALYSIS & PREDICTION
# =====

# A. Exploratory Data Analysis (EDA)
print("\n--- Generating Correlation Heatmap ---")
plt.figure(figsize=(10, 8))
correlation_matrix = df.corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title("Correlation Matrix of Patient Features vs Disease Progression")
plt.tight_layout()
plt.show()

# B. Data Preprocessing (Train/Test Split)
# Features (X) and Target (y)
X = df.drop('target_progression', axis=1)
y = df['target_progression']

# Split the data into 80% training and 20% testing
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# C. Machine Learning Model Training (Prediction)
print("Training Random Forest Regressor Model...")
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Make predictions on the unseen test data
predictions = model.predict(X_test)

# Evaluate the model
mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)

print(f"\n--- Model Evaluation ---")
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"R-squared (R2): {r2:.2f}")

# =====
# 3. PRESENT FINDINGS WITH VISUALIZATIONS
# =====

# Feature Importance Plot
feature_importances = pd.Series(model.feature_importances_, index=X.columns)
feature_importances = feature_importances.sort_values(ascending=False)

plt.figure(figsize=(10, 5))
sns.barplot(x=feature_importances.values, y=feature_importances.index, palette='viridis')
plt.title("Which Patient Features Best Predict Disease Progression?")
plt.xlabel("Importance Score")
plt.ylabel("Features")

```

```
plt.tight_layout()
plt.show()

# Actual vs Predicted Plot
plt.figure(figsize=(8, 6))
plt.scatter(y_test, predictions, alpha=0.7, color='b', edgecolor='k')
# Plot a perfectly linear line for reference (where Actual == Predicted)
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--', lw=2, label='Perfect Prediction Line')
plt.xlabel("Actual Disease Progression")
plt.ylabel("Predicted Disease Progression")
plt.title("Model Accuracy: Actual vs. Predicted Patient Outcomes")
plt.legend()
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()

print("\nProject execution complete. Please review the generated visualizations.")
```

Project Requirements Mapping

This script strictly adheres to the requested key features of the assignment:

- **Choose a dataset:** Utilizes patient records (the `sklearn` Diabetes dataset), thereby selecting the "Health" domain.
- **Perform end-to-end data analysis or prediction:** Covers EDA (Correlation Matrix), preprocessing (Train/Test splitting), and predictive modeling using a `RandomForestRegressor`. Evaluates performance using Mean Squared Error (MSE) and R-squared metrics.
- **Present findings with visualizations and conclusions:** Generates three professional charts:
 - *Correlation Heatmap:* To understand relationships among physiological variables.
 - *Feature Importance Bar Chart:* Concludes which health metrics are the strongest predictors (e.g., BMI and specific blood serum measurements).
 - *Actual vs. Predicted Scatter Plot:* Visually demonstrates model accuracy in predicting patient outcomes.